

# Interview Preparation Summary - React, JavaScript, Java, AWS

Compiled from this chat session for quick revision before interview.

## React Topics

- Hooks: useState, useEffect, useMemo, useCallback, useRef, Context API
- Virtual DOM
- Props vs State
- Controlled vs Uncontrolled components
- React.memo
- Lazy Loading and Code Splitting
- Redux Toolkit
- API integration
- Component lifecycle
- Performance optimization

## JavaScript Topics

- Closures
- Hoisting
- Event Loop
- Promises
- Async/Await
- Debounce and Throttle
- map/filter/reduce
- var let const
- === vs ==
- Array/Object coding questions

## Coding Questions

- Reverse string
- Palindrome
- Remove duplicates
- Find duplicates
- Flatten array
- Count occurrences
- Debounce implementation
- Counter app
- Search filter

- API fetch example

## **Backend Java Topics**

- Java 8 Streams
- Collections
- Exception Handling
- Multithreading basics
- Spring Boot
- REST APIs
- JPA/Hibernate
- Microservices
- JWT/Auth
- Design and architecture questions

## **AWS Topics**

- S3
- Pre-signed URLs
- SQS
- EventBridge
- API Gateway
- Lambda
- CloudWatch
- IAM
- Microservice event flow

## **Likely Project Discussion Areas**

- How API is called from React
- How data flows between FE and BE
- EventBridge to SQS flow
- S3 upload/download using pre-signed URLs
- Why hooks were chosen
- Performance optimization
- Error handling and retries

## Issue Encountered During Session

Error seen: "Content couldn't be displayed. Sorry, something went wrong. Please try again".

Possible reasons: response too large, temporary UI issue, browser/network interruption, rendering failure for large content.

## Final Revision Checklist

- JavaScript fundamentals
- React Hooks
- Redux Toolkit
- API calls and error handling
- Spring Boot basics
- EventBridge/SQS/S3 flow
- Pre-signed URLs
- Project architecture walkthrough